

indicates that the XLP is expected to change by 0.5 percent if the S&P 500 changes by 1 percent.

One observation that we found particularly striking was that despite the XLP's low beta, since the start of the index at the end of 1998, the net percentage changes in the XLP and the S&P over the entire period were almost identical. The XLP was up less in the bull markets and down less in the bear markets, but for the period as a whole, the net change was about the same. Seeing that both indexes had approximately the same net change over a long period—a period that included both the Internet boom and bust and the credit boom and bust—makes the notion that the XLP has a beta of 0.5 versus the S&P seem counterintuitive if applied to longer periods. In addition, we thought that cash flow and dividend valuations implied the potential for a 25 percent revaluation of the XLP versus the S&P. We went to an exotic option dealer and asked them to price an outperformance option that would be based on the performance of the XLP versus the S&P. What is the single measure that the dealer is going to use to price the odds that the XLP will outperform the S&P?

The beta.

Right. So with the beta equal to only 0.5, the model price for an outperformance option was very cheap. Translated into English, those inputs are saying that the XLP and S&P are likely to move in the same direction; however, the XLP will move only half as much as the S&P.

But if we had a down market, then the lower beta would imply a higher probability of outperforming—namely, it would imply that the XLP would go down less than the S&P.

That's a great point, and it is the reason why, to get the option cheaply, we had to strike the option at the current spot price. So there was a dual condition for the option to pay off: The XLP had to outperform the S&P and the S&P had to be unchanged to higher. This was essentially a conditional long beta position. It was conditional on the XLP outperforming the S&P, and it was long beta because it could only pay off in an up market.

What made you think the timing for the trade was right?